

File Compressor in C

Sample Output

Program Startup

```text id="7zwkzk"

FILE COMPRESSOR

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1. Compress File
2. Decompress File
3. View File Information
4. Exit

Enter your choice:

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## 1. Compress File

```text id="r4b22w"

Enter your choice: 1

=== Compress File ===

Enter Source File : report.txt

Enter Compressed File : report.comp

Compressing...

✓ Compression completed successfully!

Original Size : 15,360 bytes

Compressed Size : 8,192 bytes

Compression Rate: 46.67%

2. View File Information

```text id="7ml6sv" Enter your choice: 3

=== File Information ===

File Name : report.comp File Type : Compressed File File Size : 8,192 bytes Created Date : 15/06/2026  
Compression Used : Run-Length Encoding (RLE)

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## 3. Compress Another File

```text id="ftjlmw" Enter your choice: 1

=== Compress File ===

Enter Source File : notes.txt
Enter Compressed File : notes.comp

Compressing...

✓ Compression completed successfully!

Original Size : 5,120 bytes
Compressed Size : 2,764 bytes
Compression Rate: 46.02%

4. Decompress File

```text id="r7r55k" Enter your choice: 2

=== Decompress File ===

Enter Compressed File : report.comp Enter Output File : report\_restored.txt

Decompressing...

✓ Decompression completed successfully!

Output File Created: report\_restored.txt Recovered Size : 15,360 bytes

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## 5. Verify Decompressed File

```text id="8rtehk"

Enter your choice: 3

=== File Information ===

File Name : report_restored.txt
File Type : Text File
File Size : 15,360 bytes
Status : Successfully Restored

Error Handling

File Not Found

```text id="ryyx5u" Enter your choice: 1

=== Compress File ===

Enter Source File : data.txt

Error: File not found!

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### Invalid Compressed File

```text id="44n1ph"

Enter your choice: 2

=== Decompress File ===

Enter Compressed File : invalid.comp

Error: Invalid or corrupted compressed file!

Exit

```text id="m7t6mt" Enter your choice: 4

Thank you for using File Compressor.

Goodbye! ```

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## Features Demonstrated

- File Compression
  - File Decompression
  - Compression Ratio Calculation
  - File Information Display
  - File Size Comparison
  - Error Handling for Missing Files
  - Corrupted File Detection
  - Menu-Driven User Interface
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